THE UNITED REPUBLIC OF TANZANIA

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

KIBAHA PRIVATE SECONDARY SCHOOL ASSOCIATION (KIPSSA)

FORM TWO PRE-MOCK EXAMINATION

031 PHYSICS

Time: 2½ Hours **Tuesday, 27th May 2025, a.m.**

Instructions

1. This paper consists of sections A, B and C with a total of **ten (10)** questions
2. Answer **all** questions
3. Section A carries **fifteen (15)** marks, section **B seventy (70)** and section **C** carries **fifteen (15)** marks
4. Cellular phones and any unauthorized materials are not allowed in the examination room
5. Non-programmable calculators and mathematical tables may be used
6. Where necessary the following constants may be used:
   i. Acceleration due to gravity, g = $10m/s^2$
   ii. Density of water = $1.0g/cm^3$
   iii. Pie, π = 3.14

Question Number	Marks	Initial of examiners
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2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		
11.		
TOTAL		

SECTION A (15 MARKS)

1. For each of the items (i) – (x), choose the correct answer from the given alternatives and write its letter beside the item number in the answer sheet provided.

(i) A physicist is generally interested in studying the relationship between:

A. Composition and decomposition of matter

B. Matter and energy

C. Physics and Chemistry

D. Physics and energy

(ii) An engineer wanted to measure the diameter of a wire to the accuracy of three decimal places. Which of the following instruments would you recommend to be used?

A. Engineers' calipers

B. Micrometer screw gauge

C. Meter rule

D. Vernier calipers

(iii) Seypalm, weighing 65 kg, climbs up a staircase to a height of 5 m in 4seconds. The work done by Seypalm is

A. 812.5J

B. 52J

C. 3250J

D. 81.25J

(iv) What makes the last drop of tap water remain hanging for some time at the outlet of the tap soon after closing the tap

A. Capillarity

B. Osmosis

C. Surface tension

D. Viscosity

(v) You feel more pain if someone with pointed high heel shoes steps on your foot than when the same person with flat shoes steps on your foot because:

A. The flat shoes are not as heavy as the pointed ones

B. The leather of the shoes has big weight

C. Pressure exerted is greater with flat shoes

D. Pressure exerted is greater with pointed high heel shoes

(vi) Archimedes principle states that

A. The upthrust experienced by a body when partially or totally immersed in a fluid is equal to the weight of the fluid displaced.

B. Upthrust is equal to apparent loss in weight when a body is partially or totally immersed in a fluid.

C. When a body is totally or partially immersed in a fluid it displaces its own weight of the fluid in which it is immersed.

D. When a body is partially or totally immersed in a fluid, it experiences a big loss in weight.

(vii) For verifying Ohm's law, we design an electric circuit diagram in which we show the arrangement of different circuit components. We find that with respect to the resistor, the

A. Ammeter is connected in parallel and the voltmeter in series

B. Ammeter is connected in series and the voltmeter in parallel

C. Ammeter and voltmeter are both connected in series

D. Ammeter and voltmeter are both connected in parallel

(viii) Which of the following statements is not true about machine?

A. It always has efficiency less than 100%

B. Its mechanical advantage can be less than 1

C. It can also be used as speed multiplier

D. It can have mechanical advantage greater than the velocity ratio

(ix) The image which is formed on a plane mirror

A. Diminishing, real, laterally inverted.

B. The same size as the object, virtual and laterally inverted.

C. Real, magnified, same size as object.

D. Magnified, vertical, laterally inverted.

(x) The greatest pressure exerted on a smooth horizontal surface by the surface of a regular block of wood of dimensions $3cm \times 4cm \times 5cm$ and the density of 0.75g/cm^3 is:

A. 225N/m^2

B. 300 N/m^2

C. 150 N/m^2

D. 375 N/m^2

i	ii	iii	iv	v	vi	vii	viii	ix	x

2. Match the items in list A with the items in List B and answer in the space provided (05 marks)

List A	List B
(i) Geothermal energy (ii) Solar energy (iii) Wind energy (iv) Sea wave energy (v) Water energy	A. Energy from the sun B. Energy from firewood C. Energy from coal D. Hydroelectric energy E. Energy from the nuclear of the atom F. Energy from hot rocks underground G. Energy from fossils H. Energy from charcoal I. Air current energy J. Energy from batteries K. Tidal energy

i	ii	iii	iv	v

SECTION B (70 MARKS)

3. (a) (i) Explain why a ship floats while a coin sinks in sea water. Give three points.

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(ii) Draw a well labeled diagram of a hydrometer

(b) An object was weighed in space and recorded as 18N and then immersed in water and it experienced an upward force of 2N; estimate the following:

(i). Its real weight

(ii) The density of the object in (kg/m^3)

4. (a) Suppose you find a man along the road pushing a motorcycle and the motorcycle moves, but when the same man is pushing a car, he fails to move it. Briefly explain why does the man fail to push the car.

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(b) When a stone is thrown on a glass window, smashes the glass window pan to pieces, but a bullet from the gun passes through making a clean hole; explain why.

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5. (a) One vehicle, which was travelling from Juba to Gulu, was seen carrying heavy goods on its roof top and some of its passengers in the vehicle were standing. Discuss why the vehicle is likely to topple if it negotiates a corner at high speed

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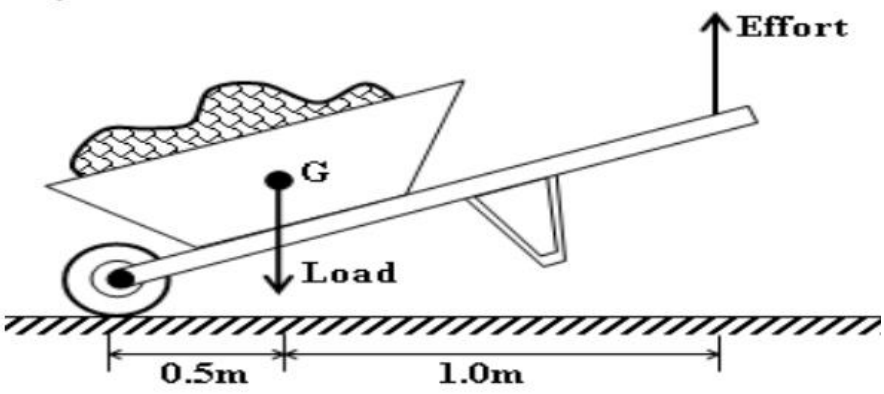
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(b) A wheelbarrow of length 1.5 m has the centre of gravity of length 0.5 m from the wheel when loaded with 50 kg as shown below. If the mass of the hand cart is 10 kg, find the effort needed to lift the wheelbarrow.



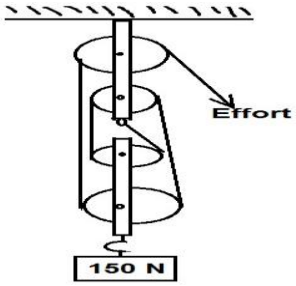
6. (a) Suppose you want to derive the relationship between displacement and time, sketch displacement-time graphs for a body at rest and for those whose rate of change of displacement with time (velocity) is not constant.

(b) A tennis ball is dropped on to the floor from a height of 10 m. It rebounds to a height of 2.5 m. If the ball is in contact with the floor for 0.01 s, what is the average acceleration during the contact? Take g = 10 m/s^2

7. (a) You have been provided with different machines with mechanical advantage (MA) of 1, 2 and 4. Which type of machine are you going to use to lift the bucket of water from the well? Explain briefly

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(b) The pulley system shown below is to be used to raise a load of 150 N

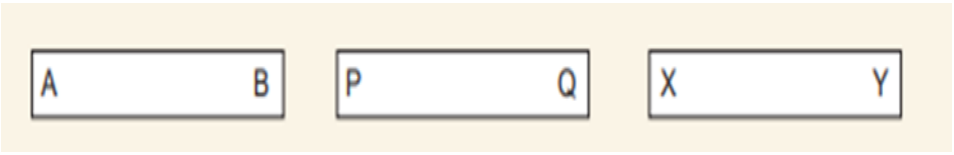


Ignoring friction and the weight of the pulleys, what force is required to raise the load at a constant speed?

8. (a) At what temperature are degree Celsius and degree Fahrenheit equal? Show involved calculations.

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(b) The magnets shown in thee figure below, pole B attracts pole P and pole Q attracts pole X



If pole Y is South Pole

(i) What is the polarity of P?

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(ii) What would happen if the following poles are brought close together?

a) Pole B and X

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b) Pole A and pole Q

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c) Pole B and pole Y

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9. (a) Describe how liquids exert pressure.

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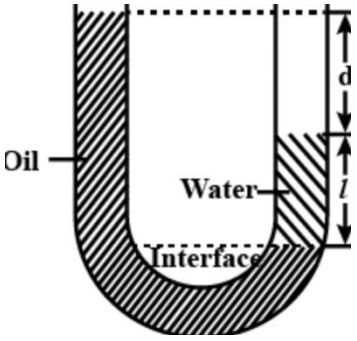
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(b) A U-tube contains two liquids in static equilibrium.

Water of density $\rho_w (= 1000 kgm^{-3})$ is in the right arm, oil of unknown density ρ is in the left arm as shown in the figure

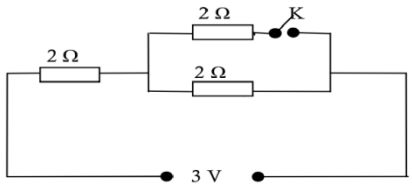


Measurement gives $l = 135\ mm$ and $d = 12.5\ mm$. Calculate the density of oil

SECTION C (15 MARKS)

10. (a) Mrs Seypalm wants someone to verify the Ohms law for her in the Physics laboratory with connecting wires, two cells, switch, fixed resistor, variable resistor, voltmeter and ammeter. As a competent student, draw the circuit showing how you will connect your device

(b) In the circuit shown in the figure below, the battery has a voltage of 3 V



Calculate the total current and current flowing in each resistor when switch K is

(i) Open 04 marks

(ii) Closed 06 marks